MITIGATING HALLUCINATION IN FINANCIAL RETRIEVAL-AUGMENTED GENERATION VIA FINE-GRAINED KNOWLEDGE VERIFICATION

Taoye Yin*, Xinya Wu*, Haoyuan Hu, Kai Deng, Yaxin Fan, Kezun Zhang, Xinhao Chen, Feng Wang

Ant Group, Hangzhou, China

1. Motivation

- **The Problem:** Financial RAG systems rely heavily on retrieved documents, but LLMs still generate hallucinations that contradict the retrieved facts (e.g., temporal or numerical inconsistencies).
- **Existing Solutions:** RL methods typically use coarse binary rewards based on human annotations. This incurs high costs and provides unstable optimization signals.
- **Our Solution:** A Reinforcement Learning framework with **Fine-grained Knowledge Verification (RLFKV)**.

Query

Did the net profit per share of Kweichow Moutai meet expectations?

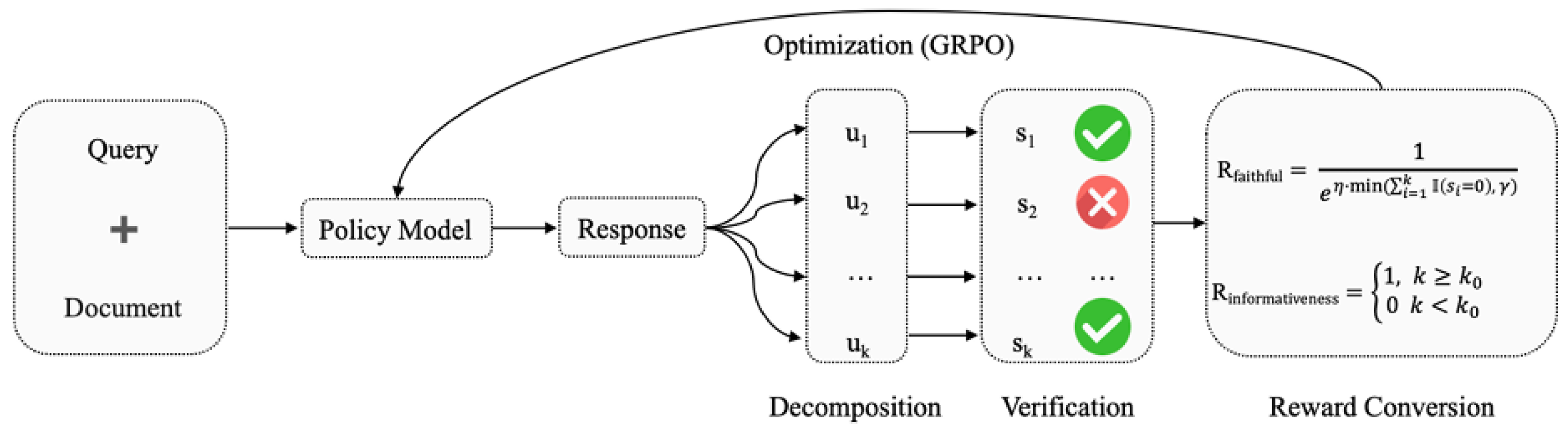
Documents

Article Title: Fundamental Analysis of Kweichow Moutai
Release Date: Thursday, May 15, 2025
Article Content: in Q1 2025, Kweichow Moutai reported…, net profit attributable to shareholders reached 26.847 billion yuan, up 11.56% year-on-year, As of March 31, 2025, the company's earnings per share (EPS) stood at 70.86 yuan, …

Response

Kweichow Moutai's net profit per share for the first quarter of 2025 indeed met and exceeded market expectations. … Additionally, data from May 15, 2025, showed that the company's basic earnings per share (EPS) stood at 70.86 yuan. …

2. RLFKV Framework



Our framework optimizes the policy model without manual reference answers through two main steps:

1. **Decomposition & Verification:** Decomposes the response into atomic knowledge units (Entity, Metric, Value, Timestamp) and verifies each against retrieved docs.
2. **Optimization (GRPO):** Uses verification results to compute fine-grained rewards.

3. Fine-Grained Rewards Formulation

We jointly maximize two reward signals to balance faithfulness and prevent reward hacking (overly concise replies).

1. Faithful Reward (r_f): Penalizes factual inconsistencies smoothly.

$$r_f = \frac{1}{e^{\eta \cdot \min(\sum_{i=1}^{k} \mathbb{I}(s_i=0), \gamma)}} \tag{1}$$

where $s_i \in \{0, 1\}$ is the verification score of unit i, and γ caps the penalty.

2. Informative Reward (r_i): Ensures informational depth matches the base model.

$$r_i = \begin{cases} 1 & \text{if } k \geq k_0 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

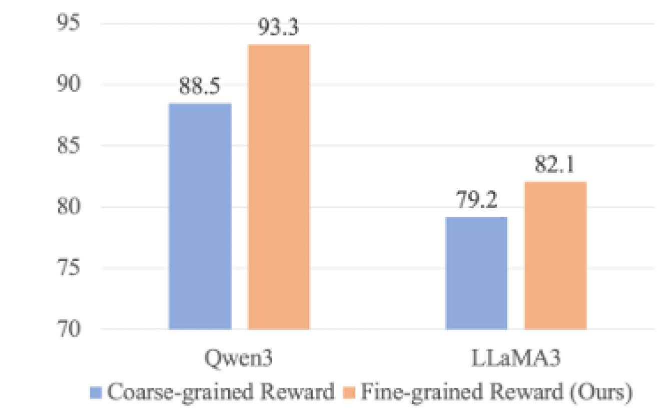
where k_0 is the number of atomic units generated by the base model π_0.

4. Main Results

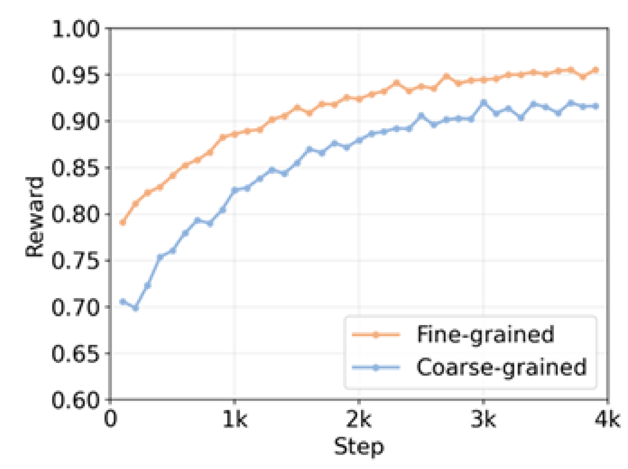
RLFKV effectively reduces hallucination while maintaining informativeness.

Table 1: Main Results

Model	FDD faith	FDD info	FDD-ANT faith	FDD-ANT info
open-source model				
DeepSeek V2 Lite	69.1	8.8	76.1	5.4
LlaMA3	80.0	11.4	80.5	6.5
Qwen3	86.5	13.4	90.2	10.8
finance model				
Xuanyuan3	57.8	7.9	64.6	5.8
Dianjin-R1	78.3	10.8	84.7	6.8
ours				
RLFKV$_{LlaMA3}$	83.6	11.7	82.1	8.1
w/o info reward	83.2	10.3	81.4	7.0
RLFKV$_{Qwen3}$	**89.5**	**13.5**	**93.3**	**12.3**
w/o info reward	89.0	12.0	91.9	11.2



(a) Performance on FDD-ANT.



(b) Reward during training.

- On FDD-ANT, $RLFKV_{Qwen3}$ improves faithfulness by **3.1 points** over the Qwen3 base model.
- Ablation studies show that removing the informative reward drops the informativeness metric significantly, validating its necessity.

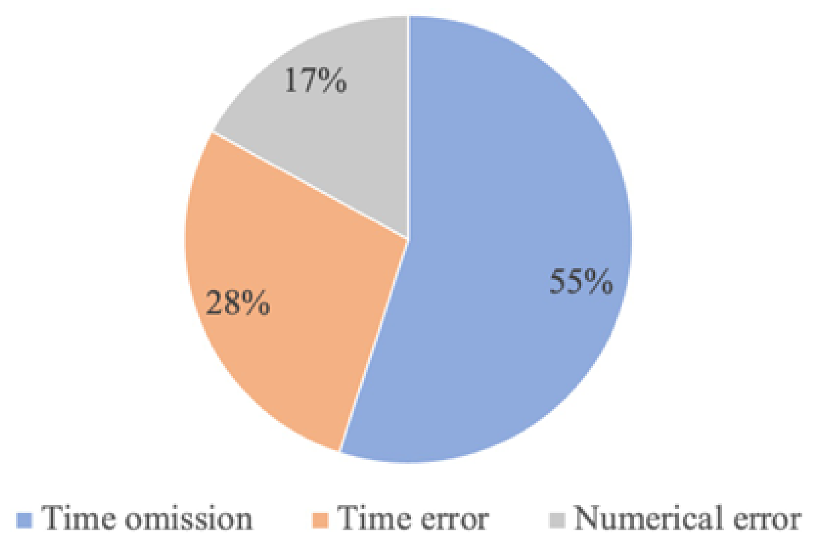
5. Reward Analysis & Error Types

Stability: Fine-grained rewards demonstrate more stable optimization and converge more rapidly (within 2k steps) compared to coarse-grained binary rewards.

Remaining Challenges: Error analysis on FDD-ANT reveals three persistent error types:

- **Time Omission (55%):** Critical temporal references are absent.
- **Time Error (28%):** Relative time expressions / calendar year conversions.
- **Numerical Error (17%):** Imprecise rounding.

Figure: Error analysis on FDD-ANT dataset.



6. Conclusion

- Proposed RLFKV, decomposing responses into verifiable atomic knowledge units.
- Achieved consistent improvements in financial RAG factual accuracy without costly human annotations.
- Released **FDD-ANT** dataset to advance real-world financial evaluation.

Scan for Code & Dataset:
github.com/antgroup/ANT-Fin-RAG

LaTeX TikZposter